

CALCULUS DONE RIGHT

FIRST EDITION

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Preface

Thank you for opening up this book. This is a book dedicated to you and those who want to learn calculus from calculus I to IV. I, myself, have taken a very windy path down the road of learning calculus, and I hope that you reading the book would not have to waste precious time like how I did. While the book assumes no calculus or precalculus knowledge, it does require knowing algebra, trigonometry and basic set theory. Also, you may feel that the precalculus section's pace is very fast, and it is intentional, this is not a precalculus book. There are many great precalculus material out there and I encourage you reading them to gain a deeper understanding on certain topics. With that said, I feel that the first section should feel manageable to most of you. In addition, while this book will cover both the theory and application of calculus, it would be more computation based instead of being very rigorous. This is not an analysis book. Hence, some parts make lack mathematical rigour but I will attach good resources to all parts of the book if I believe some other source explains certain things better than I do. Lastly, read the book thoughtfully, do not just blindly flip through the pages to not waste your precious time. I wish all readers find joy and learn something in reading the book.

Precalculus

SECTION 1

Functions

SUBSECTION 1.1

Function basics

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with these topics, you may skip to the Part II on Calculus I. For the rest of you, we will start with the idea of a function. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \quad (1.1)$$

$$\sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \quad (1.2)$$

$$y = mx + c \quad (1.3)$$

Out of the three equations, notice that 1.3 is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation "produces" a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and "produces" another value based on the input. You may have come across functions before, such as the trigonometric functions like $\sin$ and $\cos$. The value of $\sin(x)$ varies with different values of x . Of course, it is time for me to give you the definition of a function.

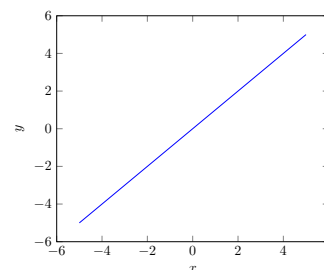


Figure 1. Graph of $y=x$.

Definition 1 A function f is a mathematical object that assigns its inputs, or domain, to its outputs, or range. The notation $f : A \rightarrow B$ means that the function has a domain A and range B .¹ The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.

¹Note that A and B are sets.

Example Here are some examples of functions:

$$f(x) = x + 1 \quad (1.4)$$

$$f(x) = \sin(x) \quad (1.5)$$

$$f(x) = x^2 - 5x + 6 \quad (1.6)$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for 1.4, it is simply the line $y = x + 1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y . Take $x = 1$ for example, $f(1) = 1 + 1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \rightarrow \mathbb{R}$. For 1.5, the function takes any real x value and maps it to $\sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just like the graph $y = \sin(x)$. Lastly, the quadratic 1.6 is another real function (i.e. it has a real domain and range). Think about what happens when $f(x) = 0$, what value of x must you plug in to make $f(x) = 0$. Well,

isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $f(x) = 0$. In this case, it would be 2 and 3. Thus, $f(2) = f(3) = 0$.

Remark

Actually, functions can be more than the usual $f(x) = \text{something}$. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such: ²

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \quad (1.7)$$

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \quad (1.8)$$

$$f(t) = (\cos(t), \sin(t)) \quad (1.9)$$

³ 1.8 is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, 1.9 does not have a single output. Instead, it has a tuple ⁴ as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

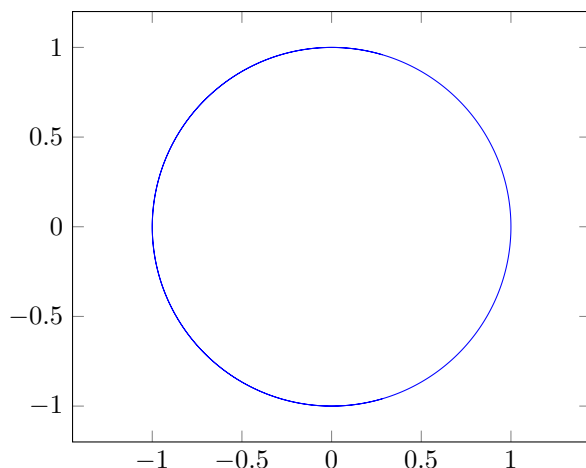


Figure 2. Plot of 1.9 for all $t \in \mathbb{R}$

²This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

³ As you will see in Part IV of the book, this is the *parametric* equation of a circle.

⁴ A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

Exercise

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:

(a) $f(x) = x^3 + 1$

(b) $f(x) = 2^x$

(c) $f(x) = \begin{cases} x^4 + 1 & \text{if } x < 2 \\ 3^x & \text{otherwise} \end{cases}$

(d) $f(x, y) = \frac{\sin(y)}{x}$

2. Given the following functions, write down $f : A \rightarrow B$ where A is the domain and B is the range (do not consider complex inputs and outputs):

(a) $f(x) = x^5$

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = \sqrt{x}$

3. Find the root(s) of the following functions:

(a) $f(x) = x^2$

(b) $f(x) = 2^x - 1$

(c) $f(x) = \sin(x)$

⁵This function has a special name, the **identity** function. Notice how its input and output are the same.

Solution For (1), simply plug in the x and y values. Hence, the answer for 1(a) is 9, 1(b) is 4, 1(c) is 9 and 1(d) is 0. On the other hand, for 2(a), notice that there are no restrictions on the domain and range, hence the answer for 2(a) is $f : \mathbb{R} \rightarrow \mathbb{R}$. However, 2(b) is not defined at $x = 0$, hence the solution is $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$. Similarly, the square root is not defined for negative numbers (not considering complex numbers), therefore the answer for 2(c) is $f : \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$. Lastly, for (3), simply find the x value such that $f(x) = 0$. The answer is as follows, 2(a) is 0, 2(b) is 1 and 2(c) is $n\pi$, $\forall n \in \mathbb{Z}$.